Web-Based Email Application

Course: Network Programming (CPAN-226-0NB)
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Agenda

Part 1: Design & implementation

Problem, tech stack, architecture, SMTP flow, data model

Part 2: Demo & evaluation

App walkthrough, limitations, conclusion & future work

Part 1: Design & implementation

How the application was built and how it works

Problem statement

Most users interact with email through polished clients like Gmail or Outlook — without understanding what happens underneath

> This project bridges that gap by building a working email client from scratch using Python's standard library

Learning objectives:

- Understand SMTP and how a mail client talks to a mail server
- Implement MIME message construction with attachments
- Apply user authentication and session management
- Persist data with a relational database

Tech stack

Layer	Technology
Language	Python 3.14
Framework	Django 6.0
Email protocol	SMTP over TLS (port 587)
Email libraries	`smtplib`, `email.mime` (stdlib — no third-party)
Database	SQLite via Django ORM
Frontend	HTML5, CSS3
Auth	`django.contrib.auth` (built-in)
Config	`python-dotenv` for credentials

> No external email libraries (e.g., `sendgrid`, `yagmail`) — all SMTP logic uses Python's stdlib to expose the protocol directly

Application architecture

The app follows Django's **MVT (Model-View-Template)** pattern

```
project/
├── emailapp/        ← project config (settings, URLs, SMTP config)
└── mailer/          ← core app
    ├── models.py    ← SentEmail database model
    ├── views.py     ← login, register, compose, sent, detail
    ├── forms.py     ← ComposeForm, RegisterForm
    └── urls.py      ← 6 routes
```

Request flow:

1. Browser sends request → Django routes to a view
2. View processes form data, calls `smtplib` to send email
3. View saves result to SQLite via ORM
4. Template renders the response back to the browser

SMTP email flow

Each outgoing email goes through these steps in `mailer/views.py`:

```
# 1. Build MIME message
msg = MIMEMultipart()
msg['From'] = EMAIL_HOST_USER
msg['To'] = to_addr
msg['Subject'] = subject
msg.attach(MIMEText(body, 'plain'))

# 2. Attach file (if uploaded) — Base64 encoded
part = MIMEBase('application', 'octet-stream')
part.set_payload(attachment_file.read())
encoders.encode_base64(part)
msg.attach(part)

# 3. Connect, upgrade to TLS, authenticate, send
with smtplib.SMTP('smtp.gmail.com', 587) as server:
    server.ehlo()
    server.starttls()          # upgrade to TLS
    server.login(user, password)
    server.sendmail(user, recipients, msg.as_string())
```

Data model

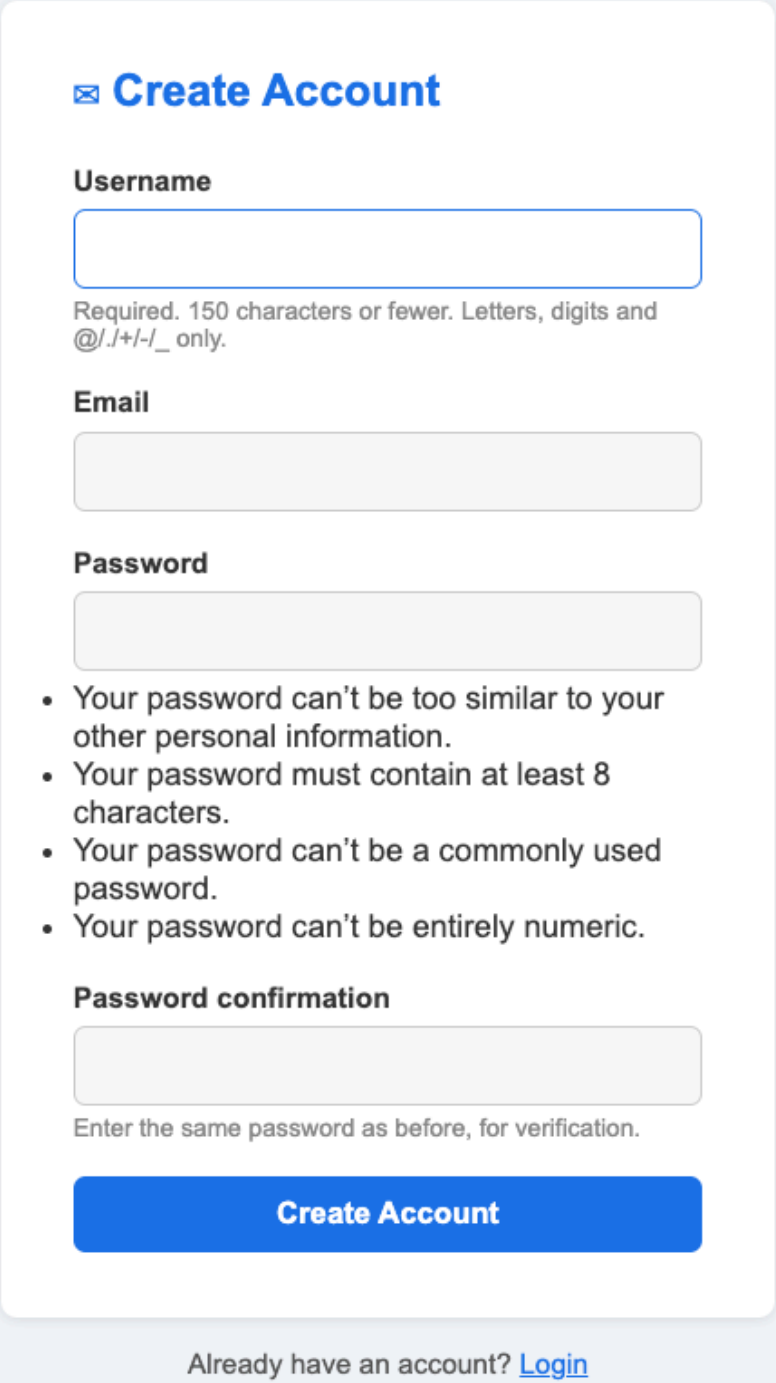
Every sent email is saved to SQLite for the logged-in user's history

```
class SentEmail(models.Model):
    sender     = models.ForeignKey(User, on_delete=models.CASCADE)
    to         = models.TextField()
    cc         = models.TextField(blank=True)
    subject    = models.CharField(max_length=255)
    body       = models.TextField()
    attachment = models.FileField(upload_to='attachments/', blank=True)
    sent_at    = models.DateTimeField(auto_now_add=True)
```

- **`auth_user`** — Django's built-in table, passwords stored as PBKDF2-SHA256 hashes
- **`mailer_sentemail`** — one row per sent email, file stored at `media/attachments/`
- Credentials kept in `.env`, excluded from version control

Part 2: Demo & evaluation

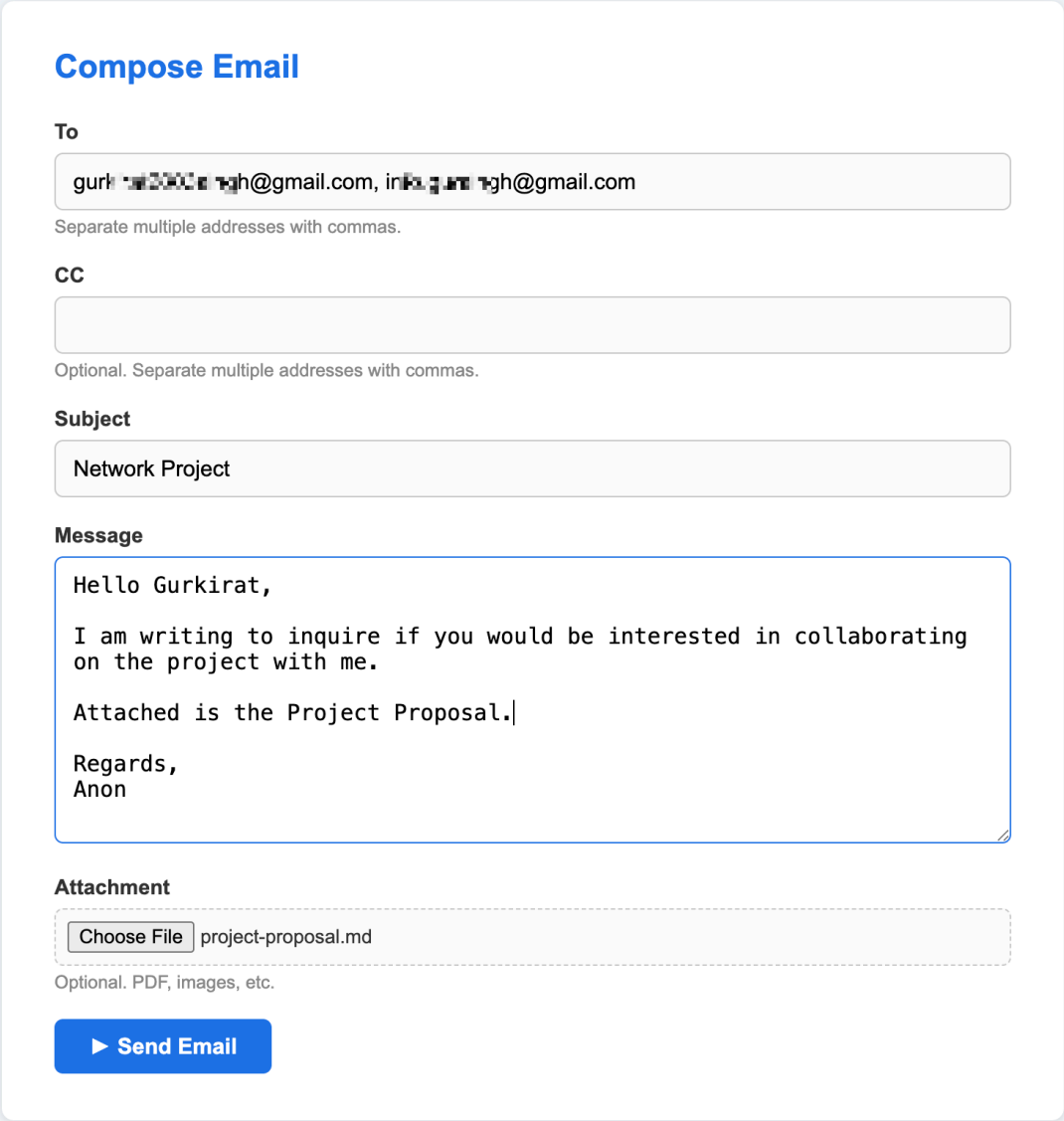
App walkthrough, limitations, and what comes next



Register & login

Register / Login

New users sign up with a username, email, and password. Django enforces password strength and hashes credentials. Invalid logins show an error — valid ones redirect to Compose.

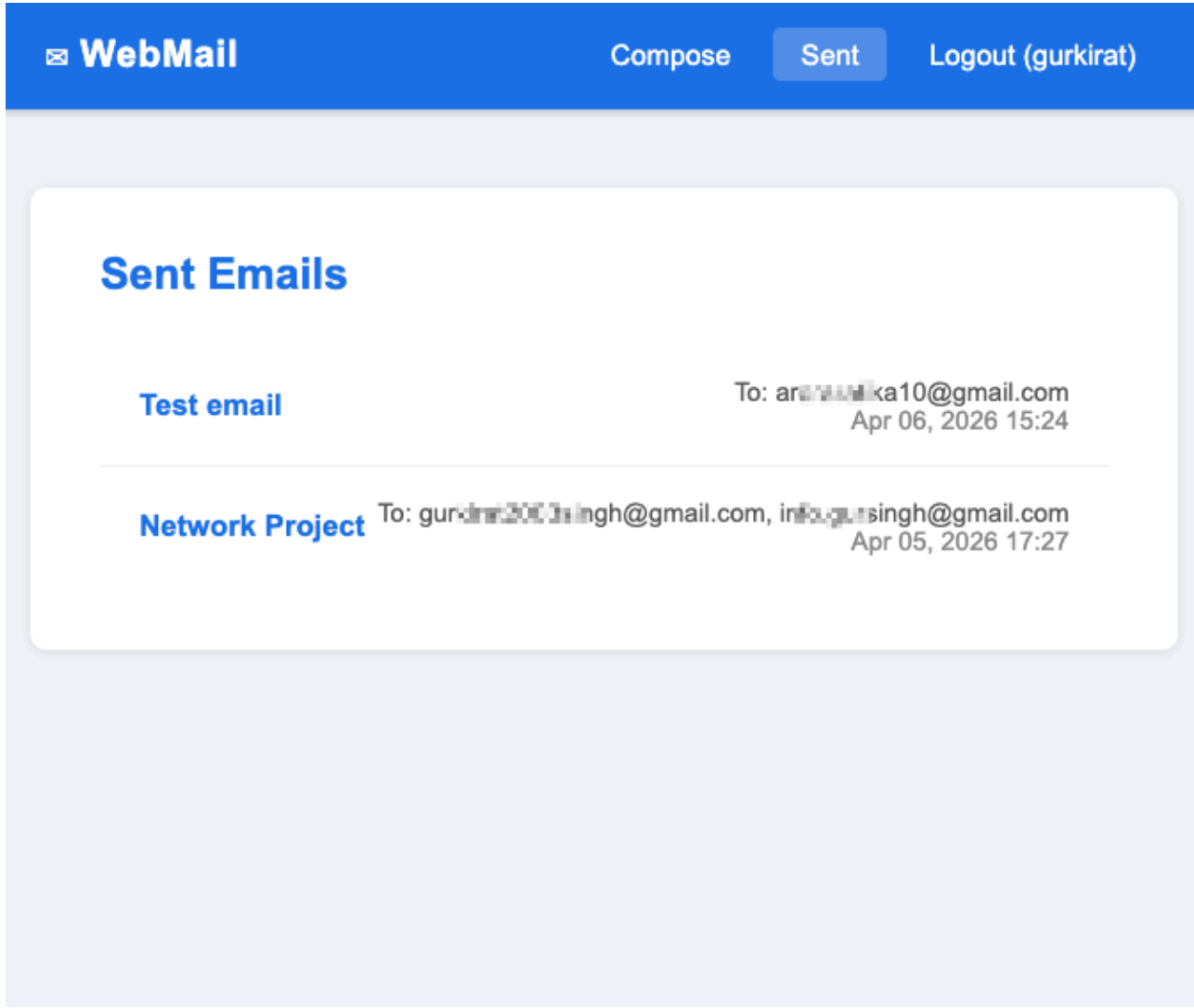


Sending an email

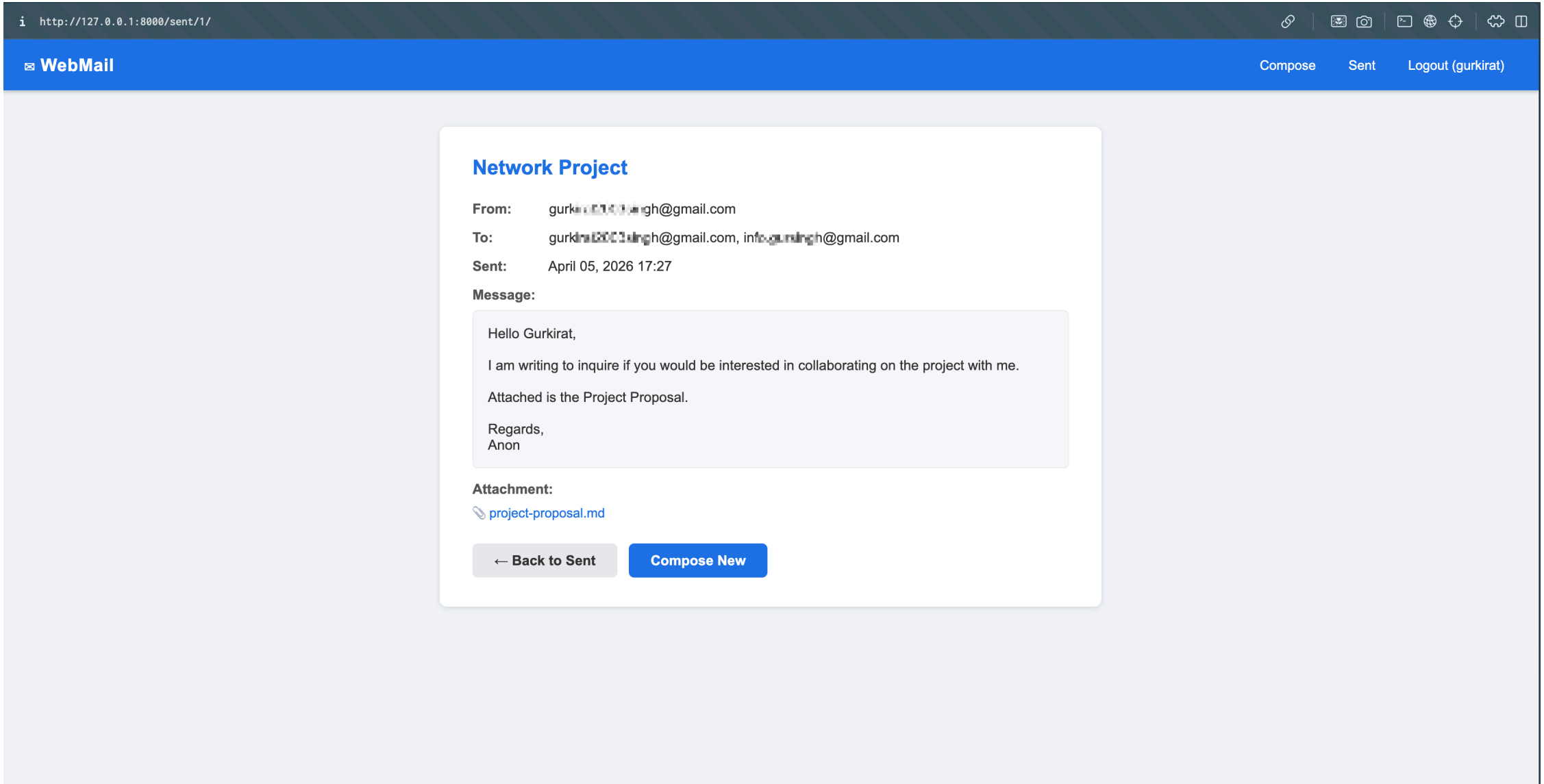
Compose — To (multi-recipient), CC, Subject, Body, optional attachment. On success a green banner confirms delivery and the user lands in the Sent folder.

Sent folder

Sent folder — lists all emails by the current user, most recent first. Clicking a row opens the full detail view with a file download link if an attachment was included.



Example Email



Limitations

- **Send-only** — no inbox; receiving email (IMAP/POP3) is not implemented
- **Shared sender identity** — all mail goes out from one configured Gmail account, not per-user addresses
- **Gmail-specific** — SMTP settings hardcoded for `smtp.gmail.com`; other providers need manual changes
- **Plain text only** — no HTML email body support
- **Development server** — not production-ready; needs Gunicorn + Nginx for real deployment
- **No search or pagination** — sent folder loads all records at once

Conclusion & future improvements

What was achieved:

- Functional web email client built with Python stdlib — no email abstraction libraries
- Direct exposure to SMTP handshake, MIME structure, and TLS
- User auth, file attachments, and database persistence all working end-to-end

Future improvements:

Improvement	Technology
Inbox (receive emails)	`imaplib` — IMAP protocol
Per-user sending	Gmail OAuth 2.0
HTML email body	TinyMCE rich-text editor
Multi-provider support	Configurable SMTP settings
Production deployment	Docker, Gunicorn, Nginx, PostgreSQL
Email scheduling	Celery + Redis task queue

References

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